

STOCK ASSESSMENT OF SOUTH ATLANTIC SWORDFISH IN 2026: EXPLORING ALTERNATIVE BIOLOGICAL PRIORS WITH JABBA MODEL

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SUMMARY

The 2026 South Atlantic swordfish assessment explored the influence of alternative biological priors on stock status estimates using the Bayesian state-space surplus production model JABBA. Informative priors for the intrinsic rate of population increase (r) and $BMSY/K$ were derived from an Age Structured Estimation Model (ASEM) under four biological scenarios combining two growth models (von Bertalanffy and Gompertz) and two steepness assumptions ($h = 0.7$ and 0.8). Each scenario was evaluated using both the assessment Joint_CPUE dataset and an expanded All_CPUEs configuration. Model performance was assessed through convergence diagnostics, residual analyses, posterior predictive checks, hindcast, retrospective analyses, and leave-one-index-out jackknife tests. Results were highly consistent across all scenarios, indicating limited sensitivity of stock status estimates to alternative biological assumptions and abundance-index configurations. Posterior estimates consistently indicated biomass above $BMSY$ and fishing mortality below $FMSY$ in the terminal year, supporting the robustness of the 2026 South Atlantic swordfish assessment.

KEY WORDS

Bayesian inference; informative priors; age-structured estimation model; abundance indices; sensitivity analysis.

1. INTRODUCTION

South Atlantic swordfish is one of the most economically important stock managed by the International Commission for the Conservation of Atlantic Tunas (ICCAT). Stock assessments have historically relied on biomass dynamic models to evaluate stock status, biological reference points, and management advice. Over time, improvements in data availability have enabled the development of integrated age-structured frameworks, broadening the understanding of stock dynamics and associated uncertainty (Mourato et al., 2022).

The Just Another Bayesian Biomass Assessment (JABBA) framework (Winker et al., 2018) has become one of the most widely used biomass dynamic tools within ICCAT. JABBA is a Bayesian state-space surplus production model that combines catch and abundance index data while accounting for both process and observation uncertainty. Its application across a wide range of Atlantic tuna and billfish stocks, including Atlantic blue marlin (Mourato et al., 2018; Mourato et al., 2024), Atlantic white marlin (Mourato et al., 2019; Mourato et al., 2025), Atlantic yellowfin tuna (Sant'Ana et al., 2019; Sant'Ana et al., 2024), Atlantic bigeye tuna (Sant'Ana et al., 2021), South Atlantic albacore and Mediterranean swordfish (Winker et al., 2020), western and eastern Atlantic skipjack tuna (Sant'Ana et al., 2022a; Sant'Ana et al., 2022b), eastern and western Atlantic sailfish (Mourato et al., 2023a; 2023b), and South Atlantic swordfish (Parker et al., 2022), has established it as a reliable platform for evaluating stock status, biological reference points, and competing hypotheses regarding stock productivity.

A key feature of the framework is its ability to incorporate external biological information through informative prior distributions, allowing knowledge derived from more complex age-structured models to be integrated while preserving the transparency and computational efficiency characteristic of biomass dynamic approaches. This capacity proved particularly relevant in the context of South Atlantic swordfish assessments. During the 2022 ICCAT process, Parker et al. (2022) applied JABBA alongside the first Stock Synthesis assessment for the stock, developed by Mourato et al. (2022). The concurrent evaluation of biomass dynamic and age-structured approaches established a multi-model framework that has since guided the scientific evaluation of the stock and informed the development of the 2026 assessment process.

Building on this framework, biological priors for the 2026 assessment were developed using an Age Structured Estimation Model (ASEM), which synthesized alternative hypotheses regarding growth and stock-recruitment dynamics (Winker et al., 2020). The ASEM provided estimates of key biological parameters and associated uncertainty that were directly incorporated into the JABBA analyses, establishing a formal linkage between the two modelling frameworks.

The objective of this study was to evaluate how alternative biologically informed prior assumptions affect estimates of stock status and management reference points for South Atlantic swordfish. JABBA models were conditioned on biological priors derived from the 2026 ASEM and fitted to available catch and abundance index data across four scenarios defined by combinations of growth models and steepness assumptions. Sensitivity analyses including alternative CPUE configurations and leave-one-index-out jackknife procedures were carried out to examine the influence of individual abundance indices on model outputs and to assess the robustness of stock status estimates across model configurations.

2. MATERIALS AND METHODS

2.1 Data

The analyses were based on the catch and abundance index datasets adopted during the 2026 South Atlantic Swordfish Data Preparatory Meeting and subsequently used in the stock assessment process. The dataset consisted of annual catches from 1950 to 2024 and standardized longline catch-per-unit-effort (CPUE) series developed by ICCAT Contracting Parties and reviewed during the preparatory meeting (**Figure 1**). Annual catches were used as the measure of fishery removals and corresponded to the official catch series adopted for the assessment. The catch history indicates a progressive expansion of the South Atlantic swordfish fishery from the 1960s through the mid-1990s, when

annual catches exceeded 20,000 t, followed by a decline and stabilization at lower levels in recent decades.

Two alternative CPUE datasets were evaluated to investigate the sensitivity of model results to the selection of abundance indices. The “Joint_CPUE” configuration corresponded to the primary dataset adopted for the 2026 assessment and included the following standardized abundance indices:

- **Joint LL** (1994–2024)
- **w_SPN LL** (1989–2024)
- **JPN LL1** (1962–1975)
- **JPN LL2** (1976–1990)
- **JPN LL3** (1991–2024)
- **CTP LL1** (1968–1990)

An alternative “All_CPUEs” configuration was developed to evaluate the influence of incorporating additional abundance information and included:

- **w_SPN LL** (1989–2024)
- **JPN LL1** (1962–1975)
- **JPN LL2** (1976–1990)
- **JPN LL3** (1991–2024)
- **CTP LL1** (1968–1990)
- **CTP LL2** (1991–2024)
- **BRA-URY LL** (1978–2024)
- **ZAF LL** (2000–2024)

Comparison between the “Joint_CPUE” and “All_CPUEs” configurations allowed the robustness of stock status estimates and management-related quantities to be evaluated under alternative abundance index compositions. The CPUE indices displayed contrasting temporal patterns, reflecting differences among fleets, fishing areas, and periods of operation (**Figure 1**). To reduce the influence of unrealistically small observation errors, all coefficients of variation (CVs) below 0.25 were replaced by 0.25. Values equal to or greater than 0.25 remained unchanged, while missing values were retained.

2.2 JABBA Model Configuration

Stock status was evaluated using the Just Another Bayesian Biomass Assessment (JABBA) framework, a Bayesian state-space surplus production model that estimates biomass trajectories, productivity parameters, and management reference points from catch and relative abundance data. JABBA separates process and observation uncertainty through a state-space formulation and has been widely applied in data-moderate fisheries assessments. All model configurations were fitted using the Pella-Tomlinson surplus production model. Observation error associated with abundance indices was fixed at 0.25 (*fixed.obsE* = 0.25), process error was estimated internally by the model, and annual process deviations were estimated throughout the entire time series (*proc.dev.all* = *TRUE*). The carrying-capacity prior (*K*) was specified as a lognormal distribution with a mean equal to six times the maximum observed catch and a coefficient of variation (CV) of 1. Initial depletion (ψ) was specified using a beta distribution with a mean of 0.95 and a CV of 0.05.

2.3 Biological Inputs and Intrinsic Rate of Population Increase Prior Derivation

Informative priors for the intrinsic rate of population increase (r) and relative biomass at maximum sustainable yield ($BMSY/K$) were derived from biological information generated by the Age Structured Estimation Model (ASEM) developed for the 2026 South Atlantic swordfish assessment. Alternative biological scenarios were constructed using two growth formulations, the von Bertalanffy Growth Function (VBGF) and the Gompertz model, combined with two steepness assumptions ($h = 0.7$ and $h = 0.8$). The biological inputs used in the equilibrium biomass analysis are summarized in **Table 1** and illustrated in **Figure 2**.

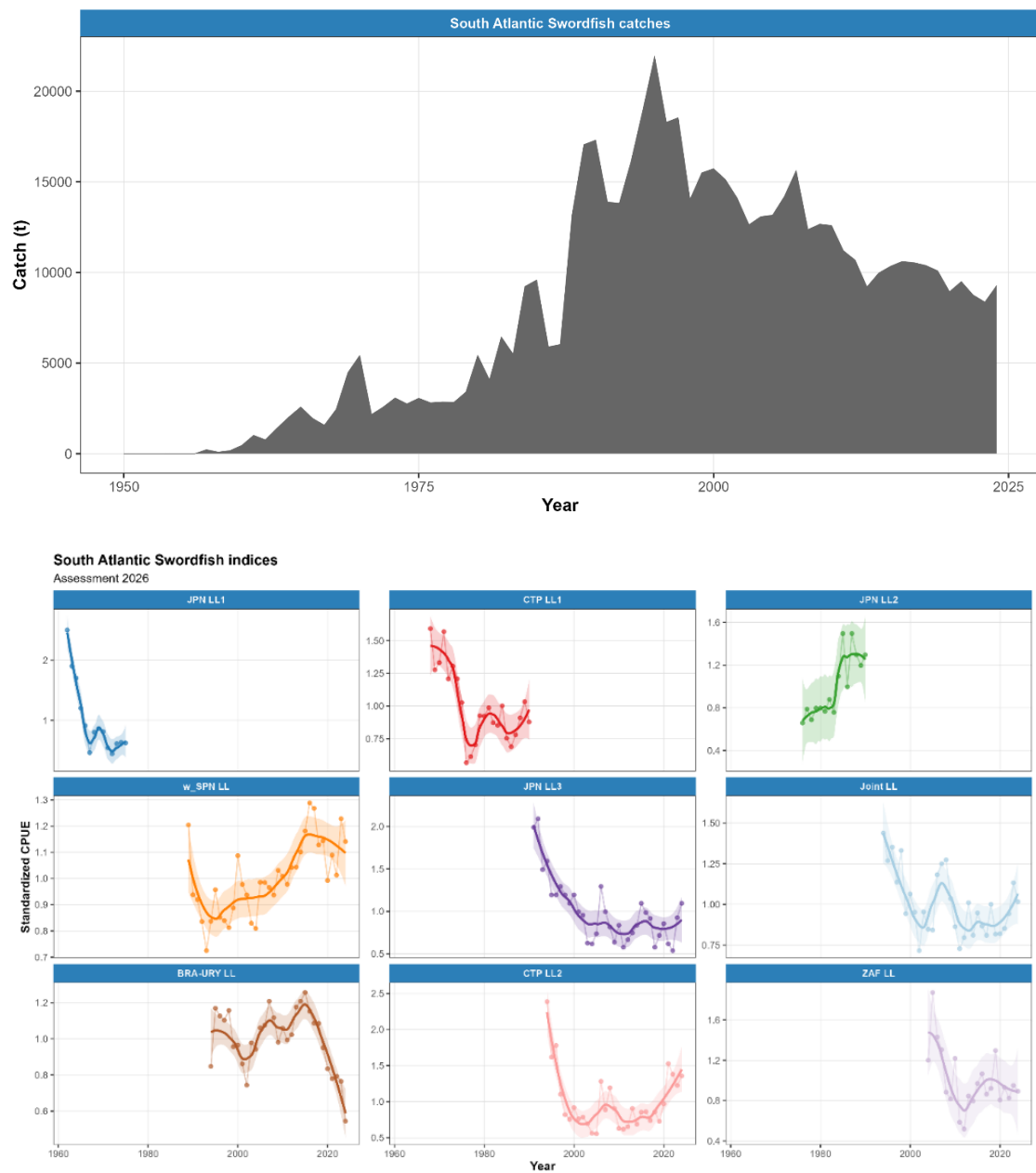


Figure 1. Catch and CPUE data used in the 2026 South Atlantic swordfish assessment. The upper panel shows annual catches (1950–2024), while the remaining panels present the standardized CPUE indices considered in the analyses. Points represent annual observations, solid lines show smoothed trends, and shaded areas indicate approximately 95% confidence intervals. The Joint_CPUE and All_CPUEs configurations were constructed from alternative subsets of these indices.

Table 1. Biological inputs used in the equilibrium biomass analysis to derive informative priors for JABBA. Growth parameters were obtained from alternative von Bertalanffy (VBGF) and Gompertz growth models, while remaining biological assumptions were common to all scenarios.

Parameter	Female VBGF	Male VBGF	Female Gompertz	Male Gompertz
L_{∞} (cm LJFL)	304.43	207.37	318.31	220.80
K (yr ⁻¹)	0.090	0.170	0.116	0.250
t_0	-3.14	-1.68	2.34	0.50
Weight-length a	1.69×10^{-6}	4.61×10^{-6}	1.69×10^{-6}	4.61×10^{-6}
Weight-length b	3.32	3.12	3.32	3.12
M	0.20	0.20	0.20	0.20
$CV(M)$	0.30	0.30	0.30	0.30
Female L50 maturity (cm)	150.7	–	150.7	–
Male L50 maturity (cm)	–	138.5	–	138.5
Selectivity L50 (cm)	119	119	119	119
Maximum age (yr)	25	25	25	25
Steepness (h)	0.7 / 0.8	0.7 / 0.8	0.7 / 0.8	0.7 / 0.8

Outputs from the ASEM were used to characterize alternative hypotheses of stock productivity and resilience. For each growth model and steepness combination, 1,000 Monte Carlo simulations incorporating uncertainty in natural mortality were conducted to estimate distributions of r and B_{MSY}/K . These distributions were subsequently used as informative priors in the assessment models (Figure 2 and Table 2). Four biological prior scenarios were evaluated:

- Gompertz_h0.7
- Gompertz_h0.8
- VBGF_h0.7
- VBGF_h0.8

Table 2. Informative priors for r and B_{MSY}/K used in the JABBA analyses. Priors were derived from age-structured model outputs under alternative growth (Gompertz and VBGF) and steepness ($h = 0.7$ and 0.8) assumptions using 1,000 Monte Carlo simulations incorporating uncertainty in natural mortality.

Scenario	Growth model	h	Mean r	CV(r)	Mean B_{MSY}/K
Gompertz_h0.7	Gompertz	0.7	0.111	0.196	0.367
Gompertz_h0.8	Gompertz	0.8	0.116	0.202	0.340
VBGF_h0.7	VBGF	0.7	0.116	0.209	0.362
VBGF_h0.8	VBGF	0.8	0.122	0.215	0.333



Figure 2. Biological inputs and informative priors used in the JABBA analyses. Upper panels show weight-at-age, length-at-age, selectivity-at-age, and female maturity-at-age relationships under Gompertz and von Bertalanffy growth models. Lower panels show the corresponding yield curves and the resulting prior distributions for the intrinsic rate of population increase (r) under alternative growth and steepness assumptions ($h = 0.7$ and $h = 0.8$). Priors were derived from 1,000 Monte Carlo simulations incorporating uncertainty in natural mortality.

2.4 Model Scenarios and Diagnostics

Each abundance-index configuration was combined with the four alternative biological prior scenarios, resulting in eight assessment models. Comparisons among model runs were used to evaluate the influence of alternative biological assumptions and abundance-index compositions on stock status estimates, biomass trajectories, productivity parameters, and management reference points. Model performance and robustness were evaluated using a suite of diagnostic analyses. Convergence of Markov chain Monte Carlo (MCMC) simulations was assessed through visual inspection of trace plots and posterior distributions, together with standard convergence diagnostics. A suite of diagnostic analyses was conducted to evaluate model fit, predictive performance, and temporal stability across all JABBA configurations. Residual patterns were examined using runs tests applied to each abundance index, with non-significant results ($p > 0.05$) interpreted as evidence of residual independence. Posterior predictive checks (PPCs) were used to evaluate the consistency between observed and model-predicted abundance indices. PPC values between 0.25 and 0.75 were considered indicative of satisfactory model performance, reflecting adequate agreement between observations and posterior predictive distributions.

Predictive skill was evaluated using hindcast cross-validation following the approach of Kell et al. (2016, 2021). Hindcast performance was summarized using the Mean Absolute Scaled Error (MASE), where values below 1 indicate forecasts that outperform a naïve random-walk prediction. Retrospective analyses were conducted by sequentially removing terminal years of data ($n = 5$) and refitting the models. Retrospective bias was quantified using Mohn's rho (ρ) following the formulation of Hurtado-Ferro et al. (2015). Values within the interval $-0.15 < \rho < 0.20$ were considered indicative of acceptable retrospective performance.

To facilitate comparison among scenarios, an overall diagnostic score was calculated by combining the proportion of successful diagnostic outcomes across four categories: runs tests, posterior predictive checks, hindcast analyses, and retrospective analyses. Greater weight was assigned to diagnostics directly related to model predictive ability and temporal consistency. The overall score was calculated as:

$$\text{Score} = 0.15(\text{Runs}) + 0.30(\text{PPC}) + 0.30(\text{Hindcast}) + 0.25(\text{Retrospective})$$

where each component represents the percentage of diagnostic tests meeting the predefined acceptance criteria. The score was used exclusively as a comparative summary metric and was not employed as a formal model-selection criterion.

3. RESULTS

All model configurations exhibited satisfactory convergence and mixing properties. Visual inspection of MCMC trace plots indicated stable chains with no evidence of trends or poor mixing. Posterior distributions were generally unimodal and well-defined, and effective sample sizes were adequate for all key parameters. Convergence diagnostics therefore supported the reliability of posterior estimates across all model runs.

All model configurations provided a satisfactory fit to the abundance indices included in both the "Joint_CPUE" and "All_CPUEs" datasets (**Figures 4 and 5**). Predicted trajectories closely reproduced the major temporal patterns observed in the CPUE series, including the historical declines evident in the Japanese longline indices, the decline and stabilization observed in Joint LL and w_SPN LL, and the trends represented by the additional BRA-URY LL, ZAF LL, and CTP LL2 series included in the "All_CPUEs" configuration. Across all biological scenarios, fitted trajectories and associated uncertainty intervals were highly similar, indicating limited influence of alternative biological priors on model fit.

Residual diagnostics showed generally consistent patterns among model configurations (**Figures 4 and 5**). Most residuals were centered around zero and exhibited no strong temporal structure, suggesting that the models adequately captured the dominant trends in the abundance data. Runs tests identified significant residual sequences for some indices, particularly JPN LL2, JPN LL3, and

CTP LL2, whereas other series, such as CTP LL1 and ZAF LL, frequently exhibited non-significant results. The JPN LL1 index showed some variation among biological scenarios, with non-significant runs tests under the VBGF configurations but significant results under some Gompertz configurations. However, these differences were not associated with noticeable changes in fitted trajectories or overall model performance.

Comparisons between the "Joint_CPUE" and "All_CPUEs" configurations revealed very similar fits to the abundance indices and comparable residual patterns. The inclusion of additional CPUE series in the "All_CPUEs" configuration did not substantially alter model behaviour, indicating that stock status estimates were primarily driven by the common information shared among the abundance indices rather than by any single series. Overall, the fit diagnostics support the robustness of the assessment results and suggest that the alternative biological priors and abundance-index configurations produced broadly consistent representations of the available data.

Additional residual diagnostics indicated very similar model performance among biological prior scenarios and abundance-index configurations (**Figure 6**). Residuals were generally centered around zero throughout the time series, with only weak temporal patterns evident in the smoothed residual trends. Root mean squared error (RMSE) values were highly consistent among model runs, ranging from 0.219 to 0.223 for the "Joint_CPUE" configuration and from 0.239 to 0.243 for the "All_CPUEs" configuration. The "Joint_CPUE" models consistently produced slightly lower RMSE values than the "All_CPUEs" models, suggesting a marginally better overall fit to the abundance data. However, differences among growth-model and steepness assumptions within each abundance-index configuration were negligible. Smoothed residual trajectories were nearly identical across biological scenarios and showed only minor departures from zero during the early years of the time series, converging toward values close to zero in more recent periods.

Estimated annual process deviations were highly consistent among biological prior scenarios and abundance-index configurations (**Figure 7**). In both the "Joint_CPUE" and "All_CPUEs" analyses, median process-error trajectories followed nearly identical patterns through time, with substantial overlap among posterior uncertainty intervals. The largest negative process deviations were estimated during the early-to-mid 1970s, followed by a gradual recovery toward values near zero during subsequent decades. Small positive deviations were observed during the late 1980s, early 2000s, and around 2010, although these fluctuations remained modest relative to the associated uncertainty. Throughout most of the time series, estimated process deviations were centered close to zero, suggesting limited evidence for prolonged periods of unusually strong or weak stock productivity.

Collectively, the convergence diagnostics, model-fit statistics, residual analyses, and process-error estimates indicate that model performance was highly robust to both biological prior specification and abundance-index composition. The strong consistency observed among alternative scenarios provides confidence that subsequent estimates of stock status and management reference points are not unduly influenced by the alternative modelling assumptions evaluated in this study.

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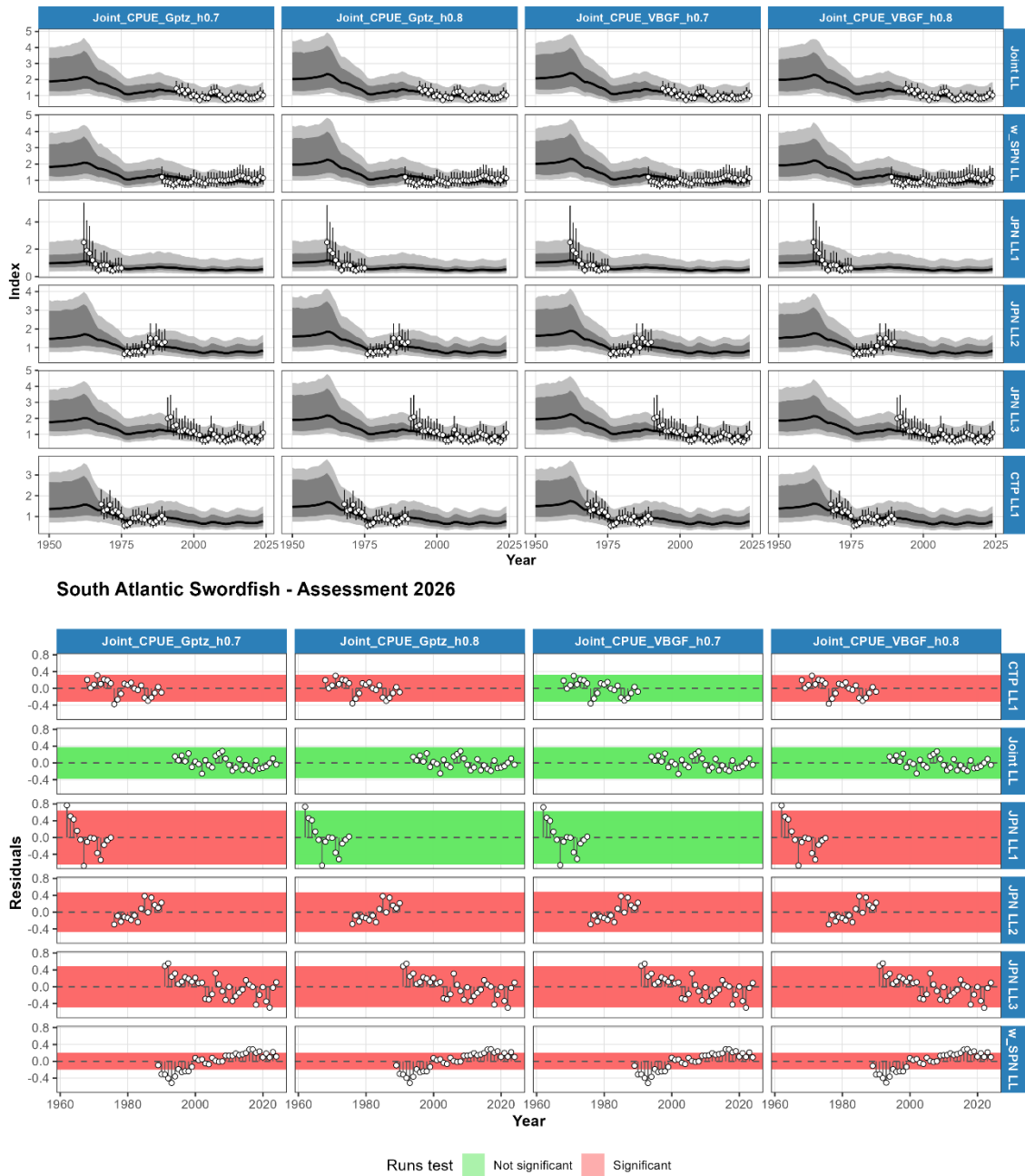
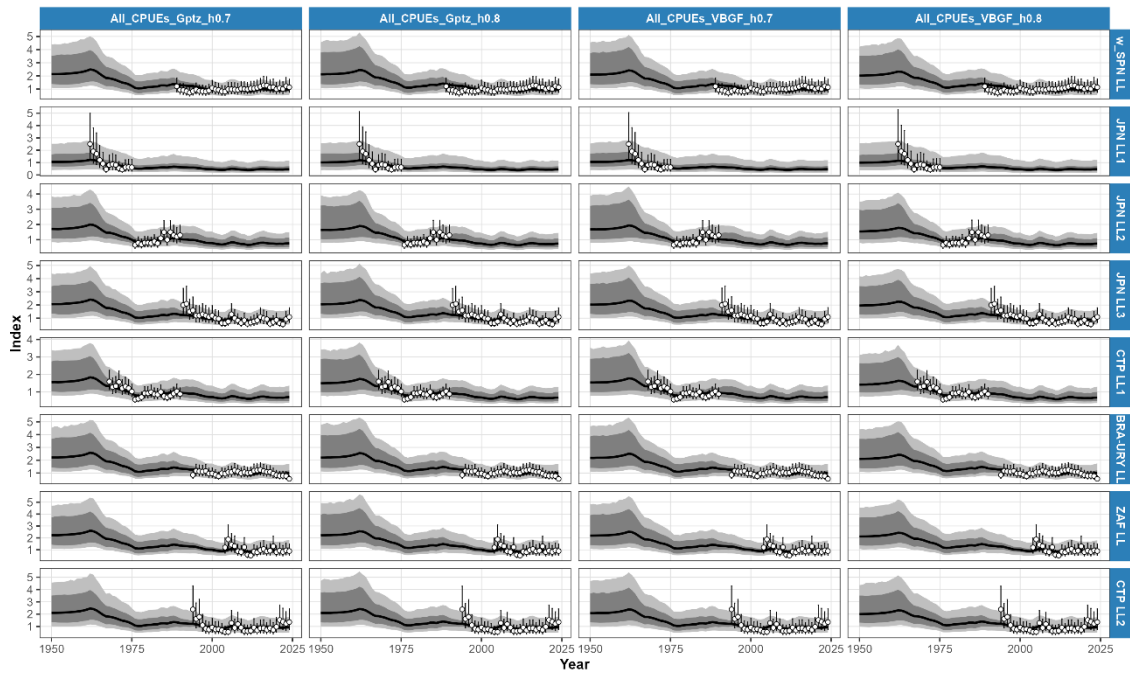


Figure 4. Model fit and residual diagnostics for the Joint_CPUE configurations under alternative biological prior scenarios. Upper panels show observed standardized CPUE indices (points with error bars) and model-predicted trajectories (solid lines) with associated uncertainty intervals (shaded areas). Lower panels show standardized residuals through time, with green and red backgrounds indicating non-significant and significant runs test results, respectively.

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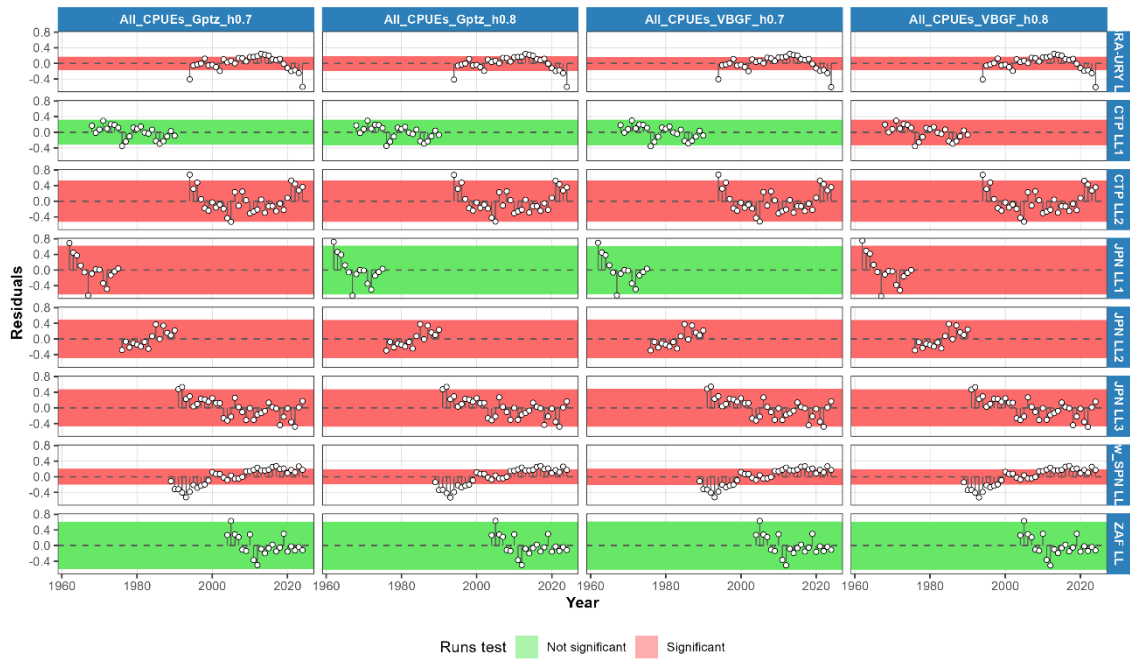
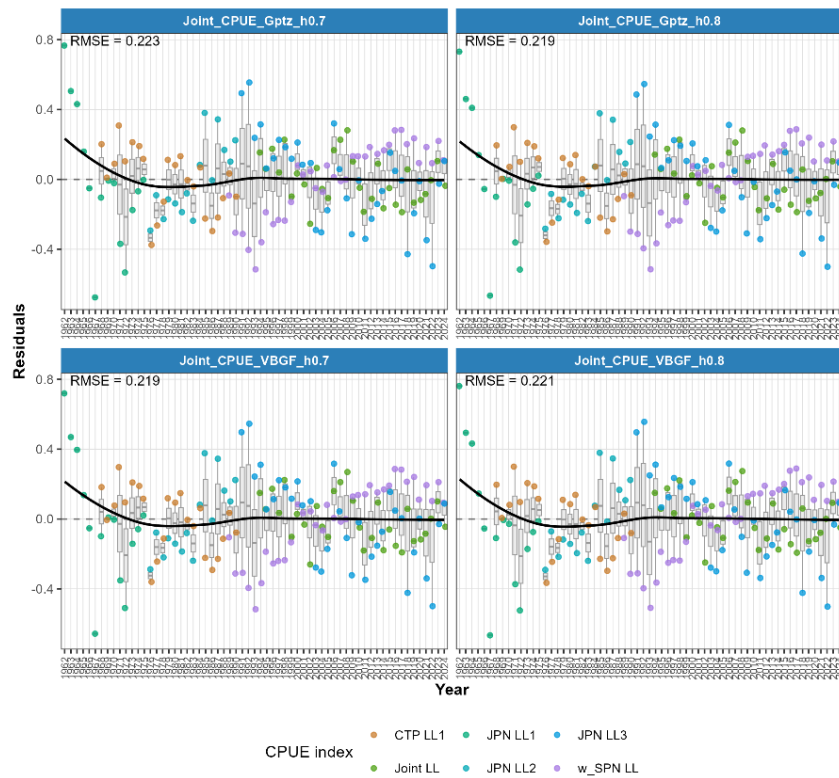


Figure 5. Model fit and residual diagnostics for the All_CPUEs configurations under alternative biological prior scenarios. Upper panels show observed standardized CPUE indices (points with error bars) and model-predicted trajectories (solid lines) with associated uncertainty intervals (shaded areas). Lower panels show standardized residuals through time, with green and red backgrounds indicating non-significant and significant runs test results, respectively. The All_CPUEs configuration includes the additional BRA-URY LL, ZAF LL, and CTP LL2 abundance indices.

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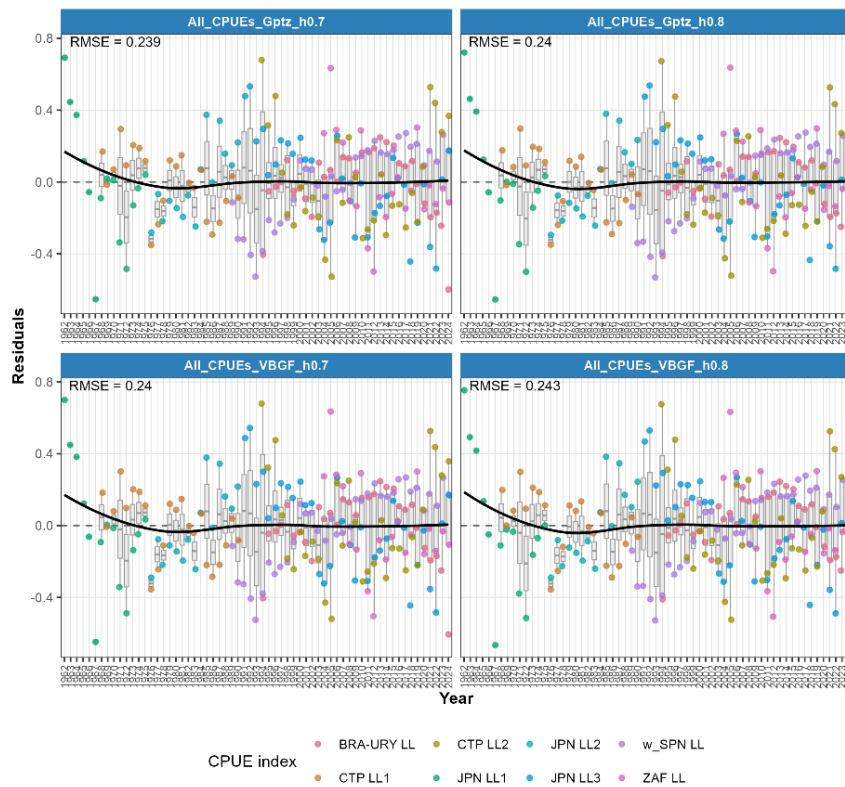


Figure 6. Residual patterns and root mean squared error (RMSE) for the Joint_CPUE and All_CPUEs model configurations under alternative biological prior scenarios. Points represent annual residuals for individual CPUE observations, coloured by abundance index. Vertical grey lines connect residuals to zero, while the black curve shows a smoothed temporal trend. RMSE values are shown for each model configuration. Similar residual distributions, smoothed trends, and RMSE values among

scenarios indicate limited sensitivity of model performance to alternative biological priors and abundance-index compositions.

Posterior distributions of the key model parameters were highly consistent among biological scenarios and abundance-index configurations (**Figure 8; Table 03**). Estimates of K , r , ψ , and σ^2 showed substantial overlap across model runs, indicating that parameter estimation was largely robust to alternative biological assumptions. A minor differences among scenarios were observed for K and r , reflecting the alternative informative priors derived from the biological analyses, while for the rest of parameters, it is observed a high overlapping among the posterior distributions across all scenarios. Scenarios based on the VBGF assumptions generally produced slightly higher posterior estimates of r than those based on the Gompertz model, whereas posterior estimates of K adjusted accordingly. Despite these differences, posterior distributions remained highly overlapping, suggesting that the abundance data provided strong information for parameter estimation. Posterior distributions of ψ were nearly identical among all model configurations, indicating a consistent reconstruction of stock conditions at the beginning of the assessment period. Likewise, estimates of σ^2 exhibited very limited variation among scenarios, supporting the conclusion that alternative biological priors had little influence on the magnitude of annual process variability estimated by the model.

Comparisons between the "Joint_CPUE" and "All_CPUEs" configurations revealed only minor differences in posterior parameter estimates. The inclusion of additional abundance indices resulted in slightly narrower posterior distributions for some parameters but did not substantially alter their central tendencies. In particular, posterior estimates of K were considerably more concentrated than the corresponding prior distributions, indicating that the available data were highly informative for this parameter. In contrast, posterior estimates of r retained some influence from the alternative biological priors, as expected for a productivity-related parameter. Overall, the strong overlap among posterior distributions indicates that parameter estimation was robust to both biological prior specification and abundance-index composition.

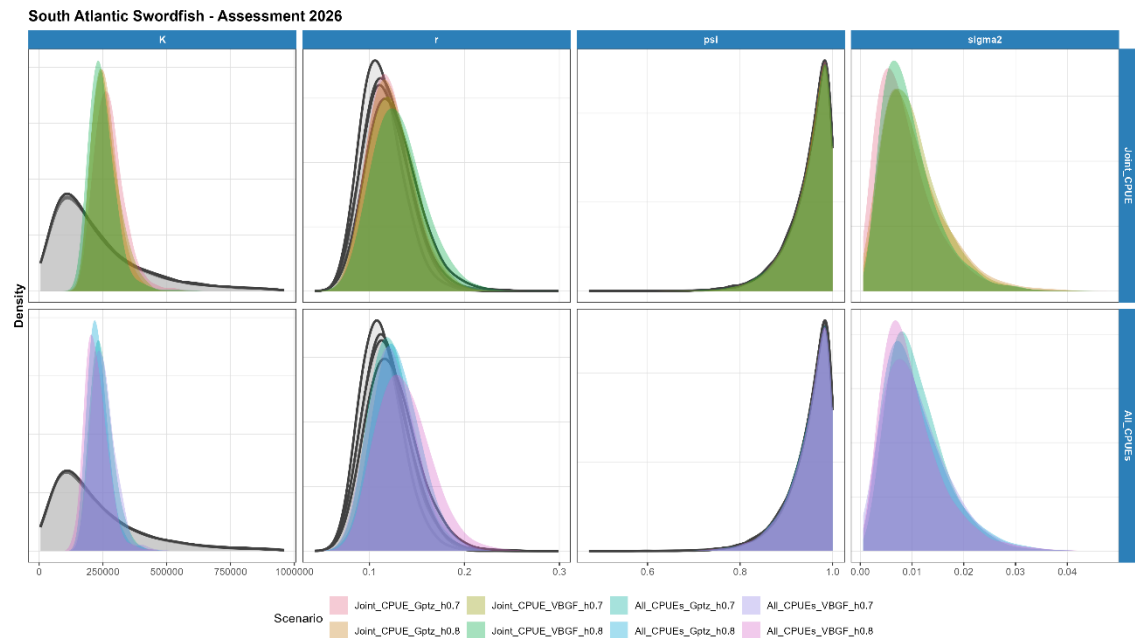


Figure 8. Posterior distributions of key model parameters for the "Joint_CPUE" and "All_CPUEs" configurations under alternative biological prior scenarios. Panels show prior (black lines) and posterior distributions for K , r , ψ , and σ^2 . Coloured densities correspond to the alternative biological scenarios. Posterior distributions were broadly consistent among scenarios, with the greatest differences observed for r , while estimates of ψ and σ^2 remained nearly identical across model configurations.

Estimated stock trajectories were highly consistent among biological prior scenarios and abundance-index configurations (**Figure 9**). In both the "Joint_CPUE" and "All_CPUEs" analyses, biomass increased slightly during the 1950s and early 1960s before declining steadily through the following decades. The most pronounced decline occurred between the mid-1960s and early 1980s, after which biomass stabilized and remained relatively constant, with only moderate interannual fluctuations. Despite differences in the underlying biological priors, trajectories of relative biomass to the virgin conditions (B/B_0), stock status (B/B_{MSY}), absolute biomass, fishing mortality, and relative fishing mortality (F/F_{MSY}) showed strong agreement among scenarios. The greatest divergence among model runs occurred during the historical period when information was limited, whereas estimates converged considerably during the period informed by abundance indices. This pattern indicates that the available data provided a strong signal regarding stock dynamics and reduced the influence of alternative biological assumptions on stock-status estimates.

All scenarios indicated that the stock remained above B_{MSY} throughout the assessment period (**Figure 9**). Estimates of B/B_{MSY} declined from values above 2.5 during the early years of the fishery to values near 1.1–1.4 in recent years. Although a substantial reduction in biomass was estimated relative to unfished levels, none of the scenarios suggested that the stock was overfished in the terminal year. Fishing mortality increased progressively from the late 1960s onward, reaching its highest levels during the 1990s and early 2000s. Estimates of F/F_{MSY} exceeded 1.0 for several years during this period, indicating episodes of overfishing. However, fishing mortality subsequently declined and remained below F_{MSY} in recent years across all scenarios. Terminal estimates of F/F_{MSY} ranged approximately from 0.5 to 0.8, suggesting that overfishing was not occurring at the end of the time series. Differences between the "Joint_CPUE" and "All_CPUEs" configurations were relatively small. The "Joint_CPUE" models generally produced slightly higher estimates of biomass and stock status, whereas the "All_CPUEs" models tended to estimate somewhat lower biomass levels and slightly higher exploitation rates. Nevertheless, the overall interpretation of stock condition was unchanged among configurations. Likewise, Gompertz- and VBGF-based scenarios generated nearly identical temporal patterns, with only modest differences in scale. Surplus-production curves were also highly consistent among scenarios (**Figure 9**). Alternative biological priors resulted in small differences in estimates of carrying capacity and stock productivity, reflected by shifts in the location of B_{MSY} and F_{MSY} . However, the overall shape of the production curves remained similar, indicating comparable representations of stock productivity across the alternative biological assumptions evaluated in this study.

Retrospective analyses indicated a high degree of stability in stock-status estimates across both abundance-index configurations and all biological prior scenarios (**Figures 10 and 11**). Sequential removal of terminal years produced relatively small changes in biomass, fishing mortality, and stock-status trajectories, while preserving the overall historical patterns inferred by the assessment. In all scenarios, retrospective peels consistently reproduced the decline in biomass from the 1960s through the 1980s, the subsequent stabilization of stock status, and the increase and later decline in fishing mortality. For the "Joint_CPUE" configuration (**Figure 10**), retrospective trajectories exhibited strong overlap among peels for Biomass, B/B_0 , B/B_{MSY} , Fishing Mortality, and F/F_{MSY} . Estimated values of Mohn's rho were generally low, indicating limited retrospective bias. Biomass-related quantities showed only modest divergence near the terminal years, whereas estimates of fishing mortality and F/F_{MSY} remained highly consistent among peels. Similar patterns were observed across both Gompertz and VBGF scenarios, suggesting that retrospective behaviour was largely insensitive to the alternative biological assumptions evaluated in this study. The "All_CPUEs" configuration produced comparable results (**Figure 11**). Although slightly greater divergence among peels was evident for some biomass-related metrics, particularly under the VBGF scenarios, retrospective trajectories remained broadly consistent and Mohn's rho values remained low. Estimates of fishing mortality and F/F_{MSY} showed particularly strong agreement among peels, indicating stable estimation of exploitation status through time. As observed for the "Joint_CPUE" configuration, the overall historical trends remained unchanged regardless of the biological scenario considered.

Comparisons between the two abundance-index configurations revealed no systematic differences in retrospective performance. The magnitude of retrospective bias was generally small relative to the overall scale of variation in biomass and fishing mortality throughout the assessment period. Collectively, the low Mohn's rho values and the strong agreement among retrospective trajectories

provide evidence that recent estimates of biomass, fishing mortality, and stock status are robust to both abundance-index composition and biological prior specification.

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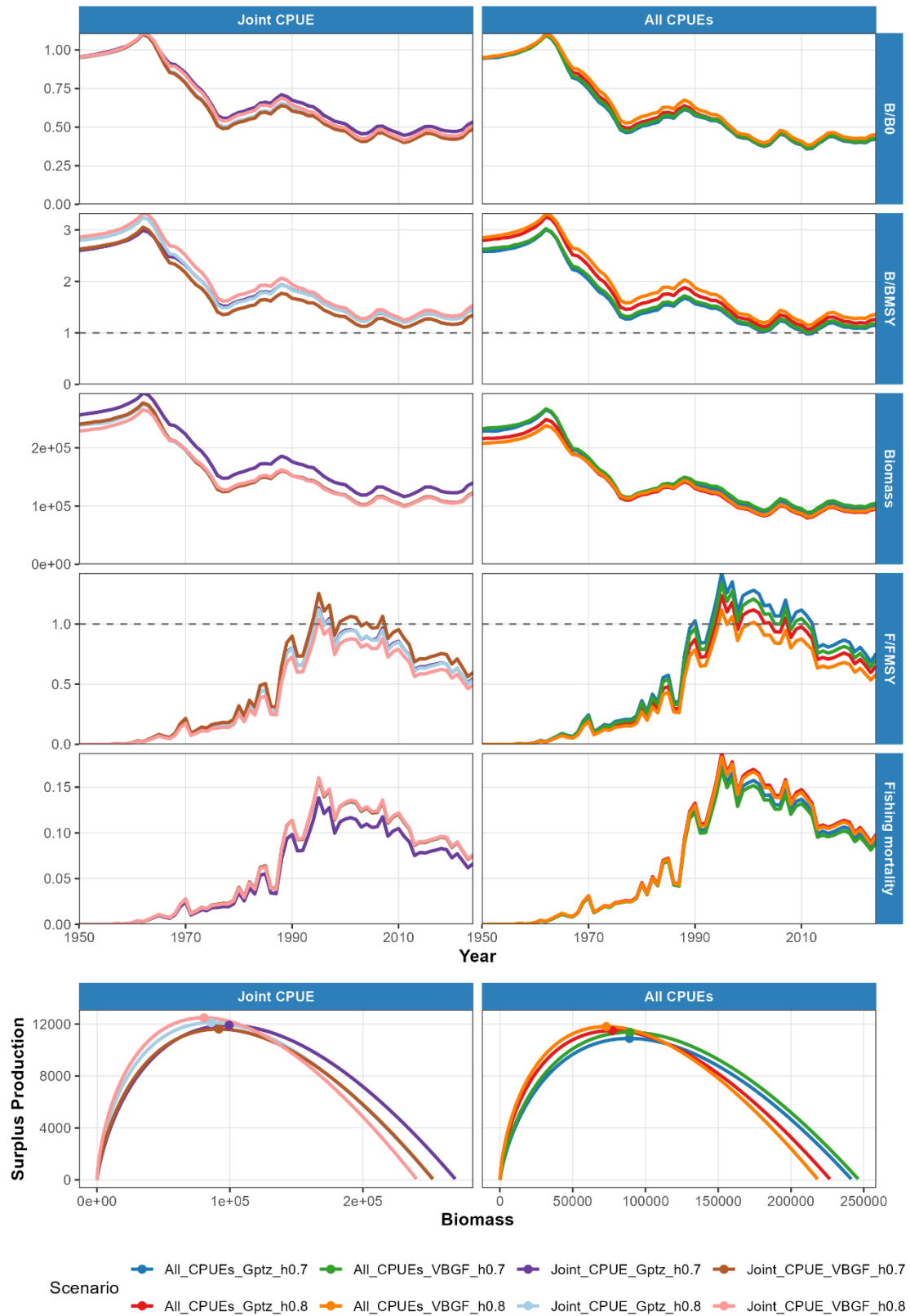


Figure 9. Historical trajectories of stock biomass, stock status, fishing mortality, and surplus production estimated under alternative biological prior scenarios for the "Joint_CPUE" and

"All_CPUEs" configurations. Panels show trajectories of B/B₀, B/B_{MSY}, biomass, F/F_{MSY}, and fishing mortality through time. Dashed horizontal lines indicate B_{MSY} and F_{MSY} reference levels. The lower panels show surplus-production curves and associated estimates of B_{MSY}. Results were broadly consistent among biological scenarios and abundance-index configurations, indicating limited sensitivity of stock-status estimates to alternative biological assumptions.

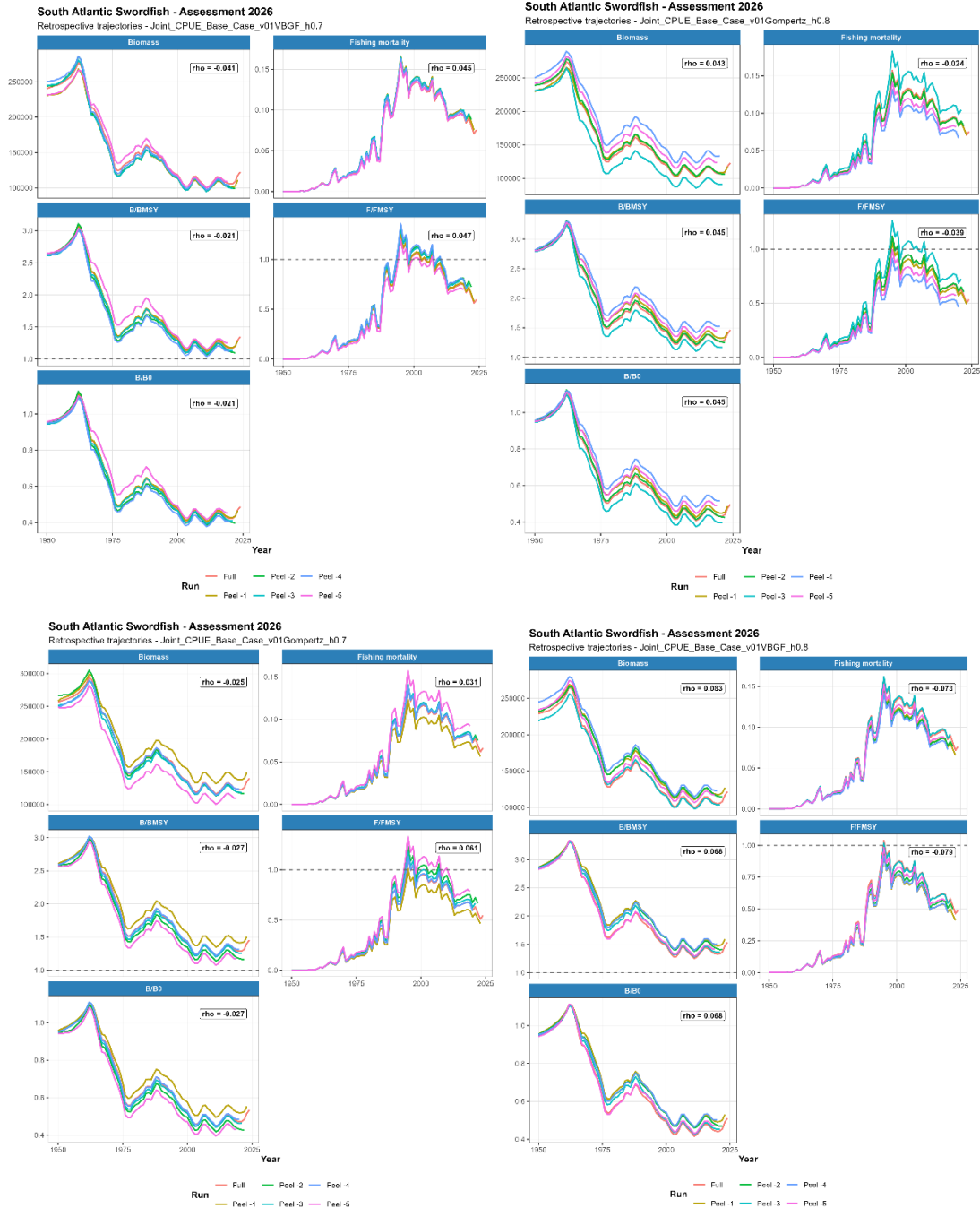


Figure 10. Retrospective analyses for the "Joint_CPUE" configuration under alternative biological prior scenarios. Panels show retrospective trajectories for Biomass, Fishing Mortality, B/B_{MSY}, F/F_{MSY}, and B/B₀ obtained by sequentially removing terminal years of data. Different colours represent successive retrospective peels. Reported values of Mohn's rho summarize the magnitude of retrospective bias for each quantity. Retrospective trajectories exhibited strong overlap and generally low Mohn's rho values, indicating limited retrospective bias.

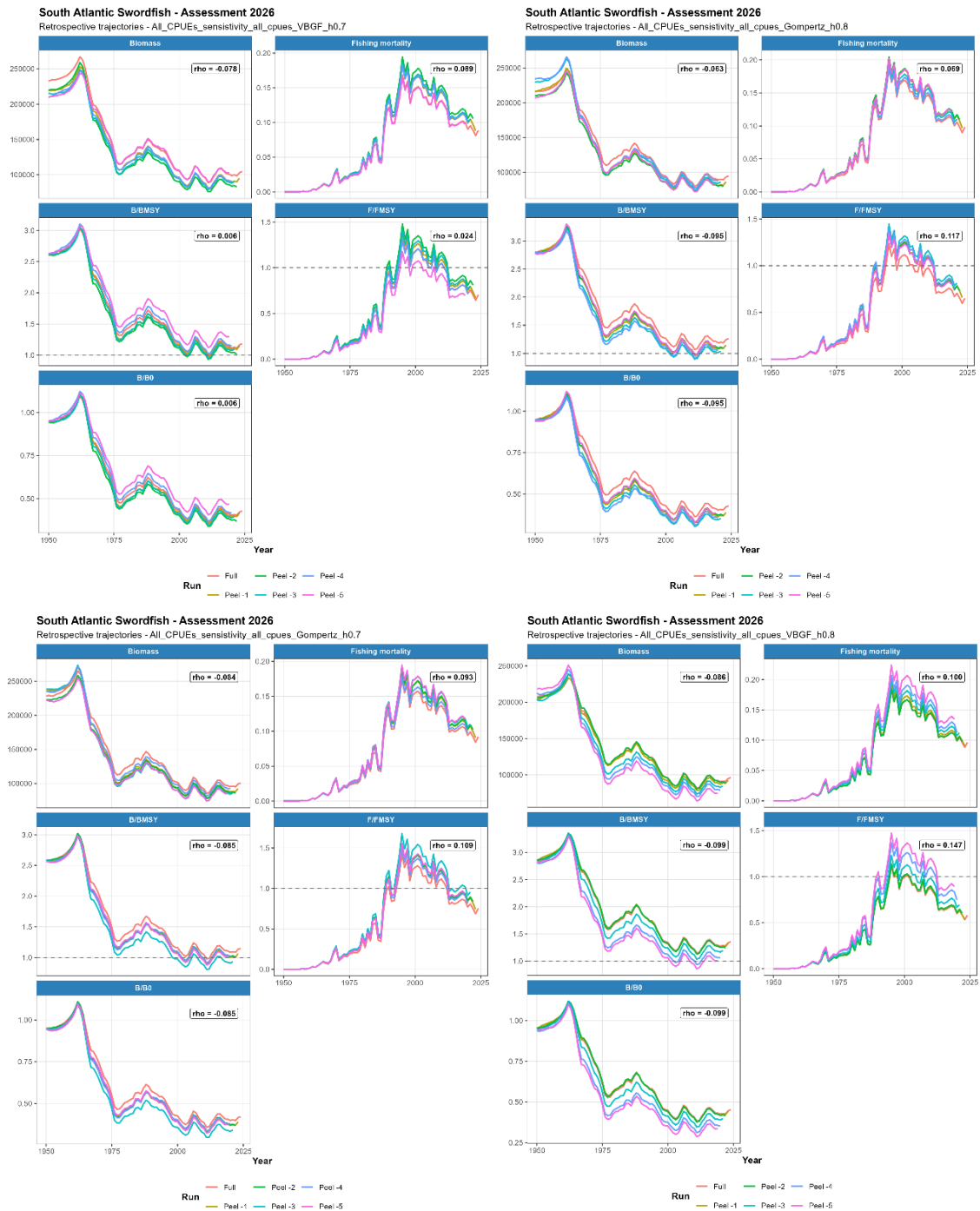


Figure 11. Retrospective analyses for the "All_CPUEs" configuration under alternative biological prior scenarios. Panels show retrospective trajectories for Biomass, Fishing Mortality, B/BMSY, F/FMSY, and B/B0 obtained by sequentially removing terminal years of data. Different colours represent successive retrospective peels. Reported values of Mohn's rho summarize the magnitude of retrospective bias for each quantity. Results were broadly consistent with those obtained for the "Joint_CPUE" configuration, supporting the robustness of stock-status estimates to alternative abundance-index compositions.

Hindcasting analyses indicated consistent predictive performance across biological scenarios and abundance-index configurations (**Figures 12 and 13**). In all cases, model predictions generated from retrospective peels closely followed the observed CPUE values withheld from the fitting process, indicating that the models were able to reproduce short-term trends in abundance after the removal of recent observations.

For the Joint_CPUE configuration (**Figure 12**), hindcast predictions generally remained within the associated uncertainty envelopes and tracked the observed values for the abundance indices included in the analysis. MASE values were relatively small and highly consistent among biological scenarios, indicating similar predictive skill under alternative growth-model and steepness assumptions. Although prediction uncertainty increased with forecast horizon, no systematic tendency toward overestimation or underestimation was evident.

Similar results were obtained for the All_CPUEs configuration (**Figure 13**). Hindcast trajectories closely reproduced the observed patterns for the abundance indices evaluated, and MASE values remained comparable among biological scenarios. The inclusion of additional abundance indices did not substantially alter predictive performance, with forecast trajectories and associated uncertainty intervals showing a high degree of consistency across model runs.

Comparisons among biological scenarios revealed only minor differences in hindcast skill. VBGF- and Gompertz-based models produced very similar forecasts, and no scenario consistently outperformed the others across all indices. Likewise, predictive performance was broadly comparable between the Joint_CPUE and All_CPUEs configurations. These results indicate that the assessment models possess adequate short-term predictive capability and provide further evidence that stock-status estimates are robust to alternative biological assumptions and abundance-index compositions.

Leave-one-index-out jackknife analyses indicated that stock-status estimates were generally robust to the exclusion of individual abundance indices in both abundance-index configurations (**Figures 14 and 15**). Across all biological scenarios, the removal of a single CPUE series produced only moderate changes in biomass trajectories, fishing mortality, stock-status indicators, and surplus-production relationships. The overall temporal patterns remained consistent regardless of the index removed, indicating that assessment outcomes were not driven by any single abundance series.

For the Joint_CPUE configuration (**Figure 14**), the largest changes were generally associated with the removal of w_SPN LL and, to a lesser extent, Joint LL. These runs tended to produce lower biomass estimates and slightly higher fishing mortality relative to the reference model. In contrast, exclusion of JPN LL1, JPN LL2, JPN LL3, or CTP LL1 typically resulted in comparatively small deviations from the reference trajectories. Despite these differences, all jackknife runs produced similar estimates of stock status in the terminal year. The All_CPUEs configuration exhibited a similar overall pattern (**Figure 15**). Although the exclusion of BRA-URY LL, ZAF LL, and w_SPN LL produced noticeable changes in biomass and fishing mortality trajectories, the resulting stock-status estimates remained broadly consistent with the reference model. Removal of CTP LL1, CTP LL2, and the Japanese longline indices generally had more limited effects on assessment outputs. The influence of individual indices was most apparent for estimates of absolute biomass, whereas relative stock-status metrics such as B/B_{MSY} and F/F_{MSY} exhibited substantially lower sensitivity.

The ranking of index influence was highly consistent among biological scenarios. Both Gompertz- and VBGF-based models identified similar responses to the removal of individual abundance series, and the magnitude of the resulting deviations remained modest relative to the overall historical variation in biomass and fishing mortality. Importantly, no jackknife run altered the general interpretation of stock condition or the relative position of the stock with respect to biological reference points. Overall, the jackknife analyses demonstrate that information supporting the assessment is distributed across multiple abundance indices rather than being dominated by a single CPUE series. The consistency of biomass, fishing mortality, and stock-status trajectories across index-removal scenarios provides additional evidence that assessment conclusions are robust to alternative abundance-index compositions and biological prior assumptions.

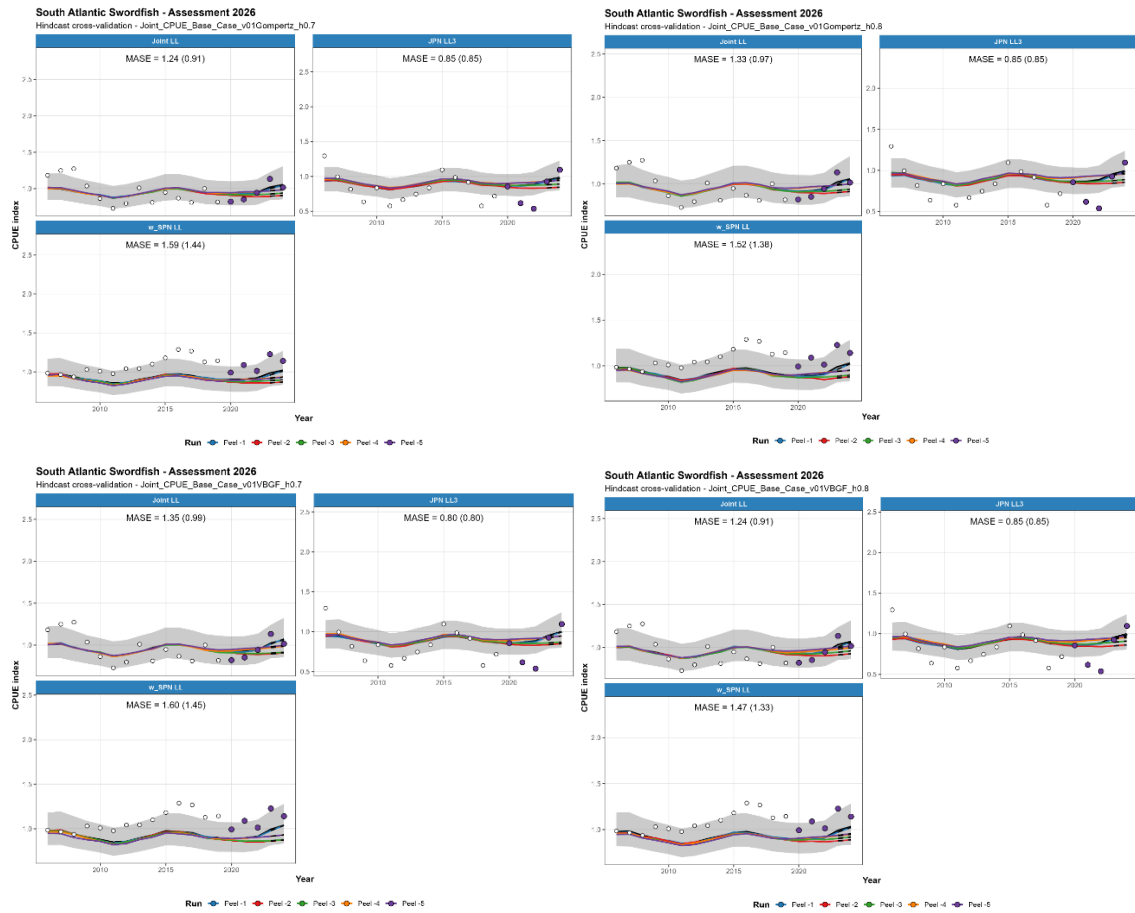


Figure 12. Hindcast evaluation for the Joint_CPUE configuration under alternative biological prior scenarios. Grey points represent observations included in model fitting, while coloured points correspond to observations withheld during successive hindcast peels. Solid coloured lines show hindcast predictions and shaded areas represent associated uncertainty intervals. Reported MAE values summarize predictive performance for each abundance index and biological scenario.

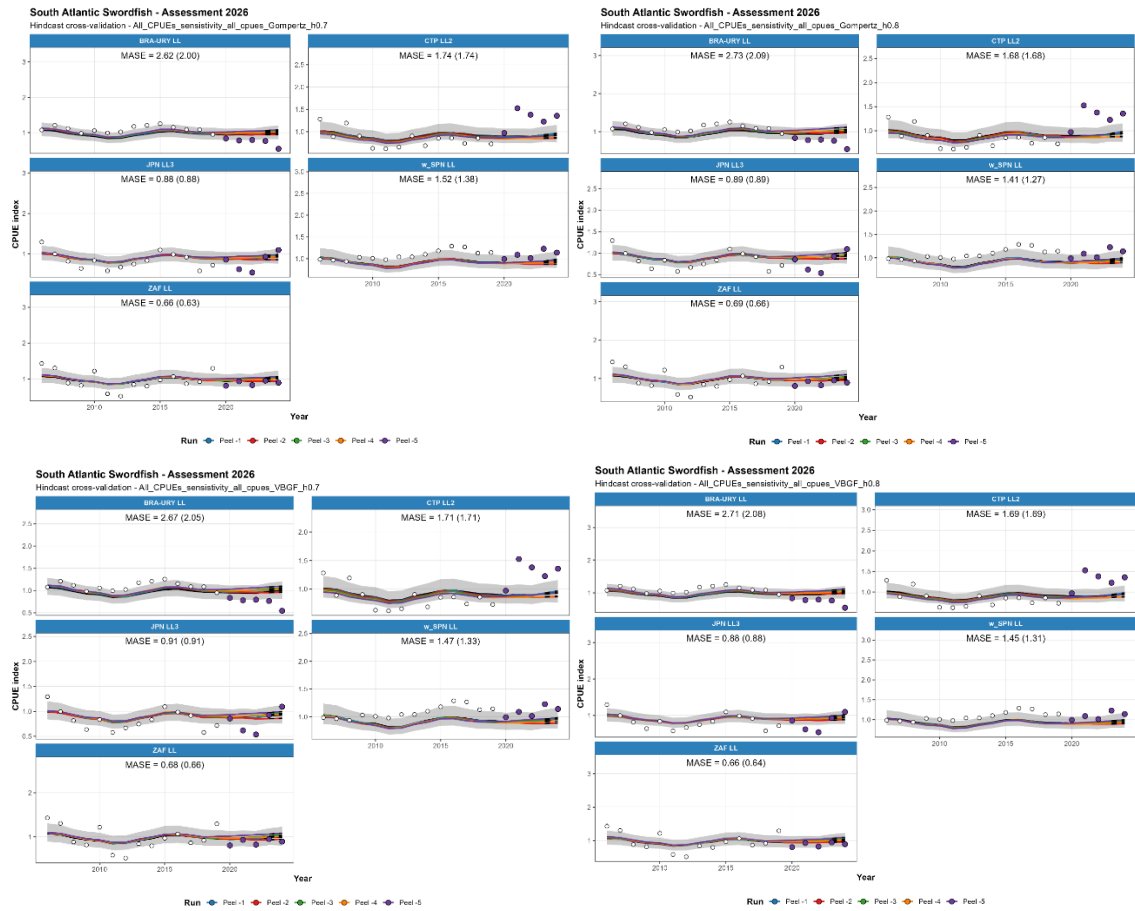


Figure 13. Hindcast evaluation for the All_CPUEs configuration under alternative biological prior scenarios. Grey points represent observations included in model fitting, while coloured points correspond to observations withheld during successive hindcast peels. Solid coloured lines show hindcast predictions and shaded areas represent associated uncertainty intervals. Reported MAE values summarize predictive performance for each abundance index and biological scenario. Predictive performance was broadly consistent among biological scenarios and abundance-index configurations.

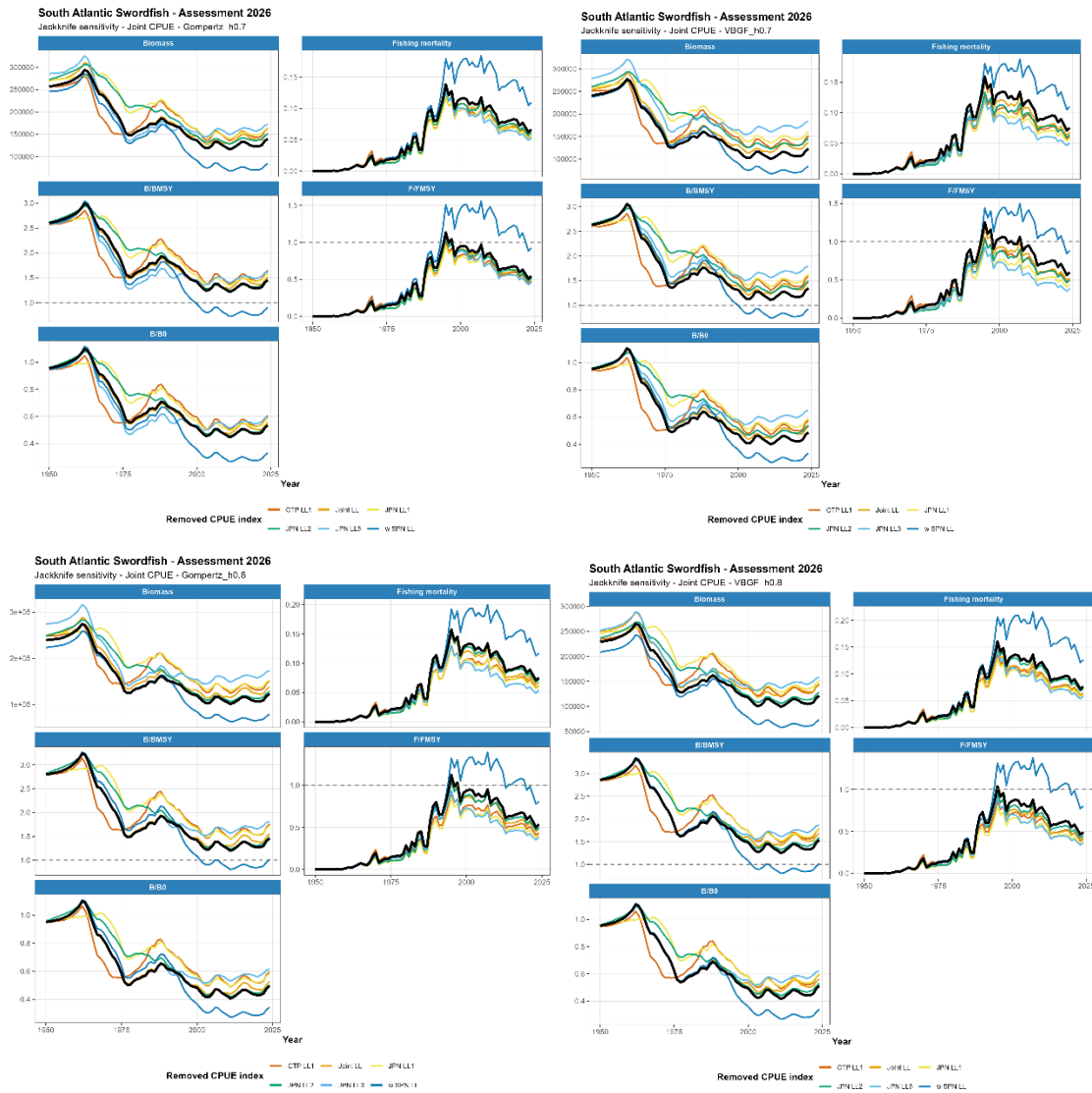


Figure 14. Leave-one-index-out jackknife analysis for the Joint_CPUE configuration under alternative biological prior scenarios. Panels show trajectories of Biomass, Fishing Mortality, B/BMSY, F/FMSY, and B/B0 obtained after sequential removal of individual abundance indices. The black line represents the reference model including all indices, while coloured lines correspond to models fitted after exclusion of a single CPUE series. Differences among trajectories quantify the relative influence of individual abundance indices on stock-status estimates. Overall, the assessment results remained broadly consistent across jackknife runs, indicating limited dependence on any single abundance index.

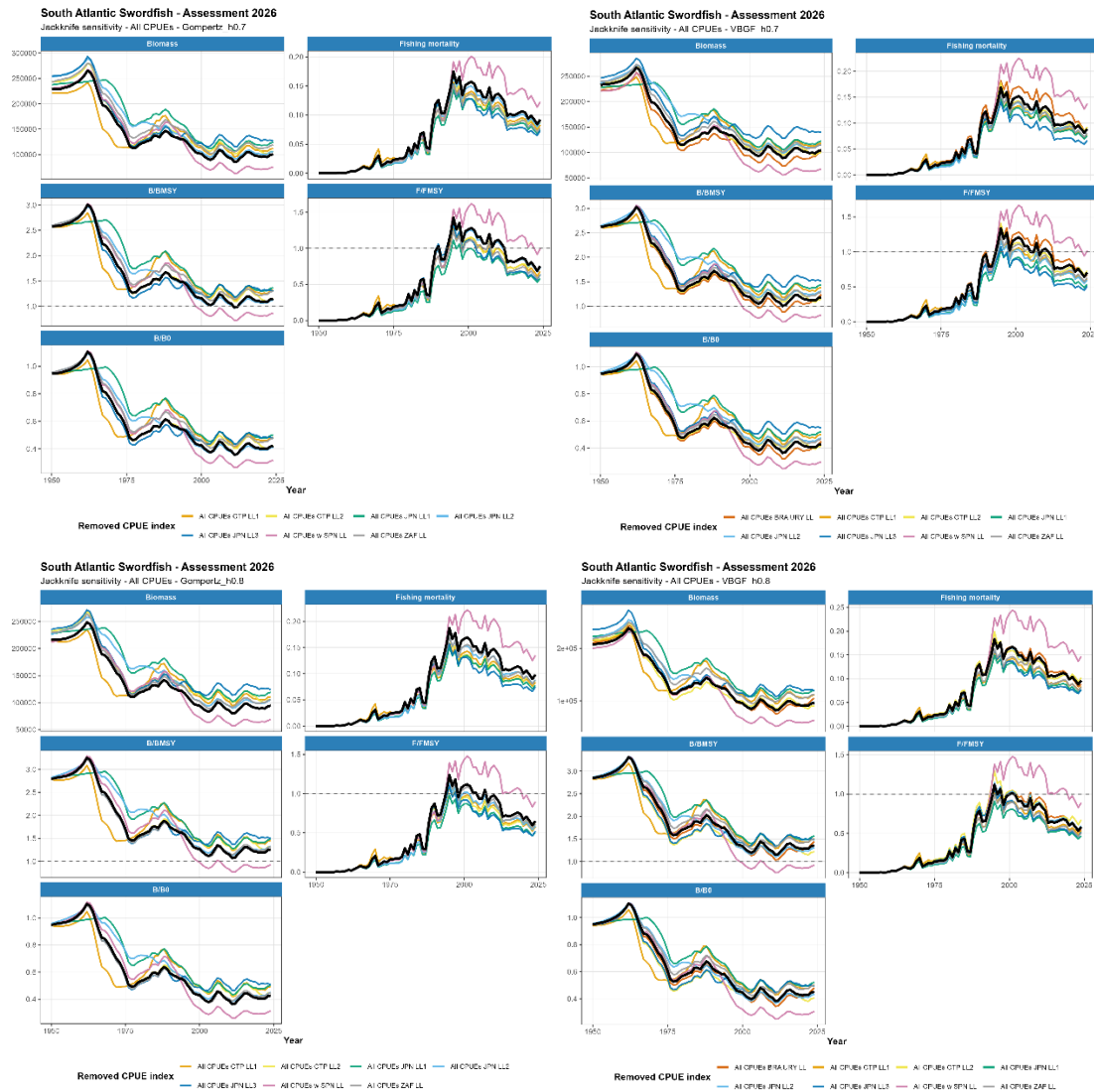


Figure 15. Leave-one-index-out jackknife analysis for the All_CPUEs configuration under alternative biological prior scenarios. Panels show trajectories of Biomass, Fishing Mortality, B/BMSY, F/FMSY, and B/B0 obtained after sequential removal of individual abundance indices. The black line represents the reference model including all indices, while coloured lines represent models fitted after exclusion of a single CPUE series. Results remained broadly consistent across index-removal scenarios, indicating limited dependence of stock-status estimates on any individual abundance index.

Diagnostic performance also showed relatively consistent results among scenarios (**Table 4**). All models successfully passed the retrospective analysis (100%), while no scenario met the hindcast criterion. Performance in the Runs Test ranged from 33.3% to 55.6%, whereas the PPC showed greater variability, with pass rates between 50% and 66.7%. Based on the weighted diagnostic score, the All_CPUEs_VBGF_h0.7 scenario achieved the highest overall performance (51.7), followed closely by All_CPUEs_Gptz_h0.8 (51.7), whereas the Joint_CPUE scenarios generally exhibited slightly lower overall scores.

Posterior estimates of the principal management quantities were generally consistent across the eight JABBA scenarios (**Table 3**). Although differences were observed among biological priors and CPUE configurations, estimated stock status remained relatively stable. Median estimates of B/BMSY were consistently above one, whereas F/FMSY remained below one for all scenarios, indicating that the stock is most likely not overfished and that overfishing is not occurring under any of the biological hypotheses evaluated.

Kobe phase plots indicated that stock status was robust to alternative biological assumptions, although uncertainty varied among scenarios (Figure 16). Across all model configurations, the largest proportion of the posterior distribution occurred within the Kobe green quadrant ($B/BMSY > 1$ and $F/FMSY < 1$). Models fitted using the Joint_CPUE configuration produced the highest posterior probabilities of the stock occurring in the green quadrant, ranging from 88.7% to 96.2%, with less than 4% of posterior samples falling within the red quadrant. In contrast, models fitted using All_CPUEs resulted in lower probabilities of the stock being in the green quadrant (70.1–88.4%) and correspondingly higher probabilities of the yellow and red quadrants, reflecting increased uncertainty associated with the inclusion of all available abundance indices.

Despite these differences, the posterior median stock status remained within the Kobe green quadrant for all eight scenarios, indicating that the overall assessment conclusions were robust to alternative biological priors and CPUE configurations. The principal effect of including all CPUE series was an increase in posterior uncertainty rather than a change in the central estimate of stock status. Moreover, differences among the four biological prior combinations were comparatively small relative to the effect of CPUE configuration, suggesting that the choice of abundance indices had a greater influence on uncertainty than the alternative growth and steepness assumptions considered in this study.

Table 03. Posterior estimates of key management quantities obtained from the eight JABBA scenarios evaluated for South Atlantic swordfish. Results are presented for the four biological prior combinations derived from the Age Structured Estimation Model (ASEM) under the Joint_CPUE and All_CPUEs configurations. For each scenario, posterior median estimates and 95% credible intervals are shown for intrinsic population growth rate (r), carrying capacity (K), biomass at MSY ($BMSY$), fishing mortality at MSY ($FMSY$), current biomass relative to $BMSY$ ($B/BMSY$), current fishing mortality relative to $FMSY$ ($F/FMSY$), and management reference points.

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		All_CPUEs - All_CPUEs_Gptz_h0.7			All_CPUEs - All_CPUEs_Gptz_h0.8			All_CPUEs - All_CPUEs_VBGF_h0.7			All_CPUEs - All_CPUEs_VBGF_h0.8		
		Median	2.5%	97.5%	Median	2.5%	97.5%	Median	2.5%	97.5%	Median	2.5%	97.5%
Kobe	K	241926.836	175063.751	355384.909	227164.003	163609.818	337755.803	246664.522	172367.402	359328.734	218669.262	153246.235	330201.930
	r	0.122	0.083	0.177	0.127	0.087	0.183	0.123	0.083	0.183	0.134	0.089	0.199
	sigma_proc	0.010	0.003	0.025	0.009	0.003	0.026	0.010	0.002	0.026	0.009	0.003	0.026
	F_MS	0.123	0.084	0.178	0.149	0.101	0.214	0.127	0.085	0.189	0.162	0.108	0.240
	B_MS	88821.612	64273.335	130476.887	77115.256	55540.547	114657.802	89271.248	62382.109	130045.959	72888.930	51081.501	110066.066
	MSY	10976.484	8298.939	14305.051	11526.305	8771.146	15578.419	11386.418	8577.656	15261.994	11885.026	9090.121	15813.513
	B1950/K	0.946	0.741	1.162	0.950	0.746	1.160	0.949	0.744	1.163	0.949	0.743	1.158
	B2024/K	0.422	0.260	0.627	0.428	0.245	0.649	0.428	0.246	0.653	0.453	0.256	0.653
B2024/BMSY	B2024/BMSY	1.149	0.709	1.707	1.260	0.723	1.911	1.162	0.679	1.804	1.358	0.767	1.958
	F2024/FMSY	0.753	0.427	1.269	0.651	0.335	1.209	0.705	0.363	1.276	0.582	0.325	1.109
		Joint_CPUE - Joint_CPUE_Gptz_h0.7			Joint_CPUE - Joint_CPUE_Gptz_h0.8			Joint_CPUE - Joint_CPUE_VBGF_h0.7			Joint_CPUE - Joint_CPUE_VBGF_h0.8		
		Median	2.5%	97.5%	Median	2.5%	97.5%	Median	2.5%	97.5%	Median	2.5%	97.5%
Kobe	K	270037.158	185402.209	419778.454	252989.656	176883.577	375241.017	253167.420	180977.061	380030.330	240813.997	164822.715	379152.635
	r	0.120	0.082	0.173	0.121	0.083	0.177	0.123	0.082	0.179	0.128	0.084	0.193
	sigma_proc	0.007	0.002	0.023	0.009	0.002	0.026	0.009	0.003	0.024	0.008	0.003	0.023
	F_MS	0.120	0.083	0.174	0.141	0.097	0.207	0.127	0.084	0.185	0.155	0.101	0.234
	B_MS	99142.105	68069.022	154118.492	85882.271	60046.584	127382.889	91624.735	65498.069	141157.285	80270.425	54940.284	126382.783
	MSY	11895.729	8655.871	17014.636	12069.441	8935.616	17066.085	11710.830	8750.730	15866.370	12430.669	9287.546	17943.521
	B1950/K	0.951	0.755	1.149	0.951	0.741	1.159	0.952	0.746	1.160	0.951	0.746	1.159
	B2024/K	0.531	0.306	0.753	0.496	0.275	0.735	0.488	0.291	0.692	0.512	0.320	0.765
B2024/BMSY	B2024/BMSY	1.454	0.834	2.051	1.461	0.811	2.165	1.348	0.805	1.912	1.535	0.981	2.296
	F2024/FMSY	0.547	0.294	1.075	0.535	0.275	1.056	0.597	0.335	1.068	0.491	0.245	0.889

Table 04. Summary of diagnostic performance for the eight JABBA scenarios. Values represent the percentage of diagnostics successfully passed for the Runs Test, Posterior Predictive Check (PPC), Hindcast and Retrospective analyses. The weighted score corresponds to the combined diagnostic performance used to compare model configurations.

Scenario	Runs pass (%)	PPC pass (%)	Hindcast pass (%)	Retrospective pass (%)	Weighted score
Joint_CPUE_Gptz_h0.7	33.3	50.0	33.3	100	55.0
Joint_CPUE_Gptz_h0.8	33.3	50.0	0.0	100	45.0
Joint_CPUE_VBGF_h0.7	50.0	50.0	33.3	100	57.5
Joint_CPUE_VBGF_h0.8	33.3	50.0	0.0	100	45.0

All_CPUEs_Gptz_h0.7	33.3	66.7	0.0	100	50.0
All_CPUEs_Gptz_h0.8	44.4	66.7	0.0	100	51.7
All_CPUEs_VBGF_h0.7	44.4	66.7	0.0	100	51.7
All_CPUEs_VBGF_h0.8	33.3	66.7	0.0	100	50.0

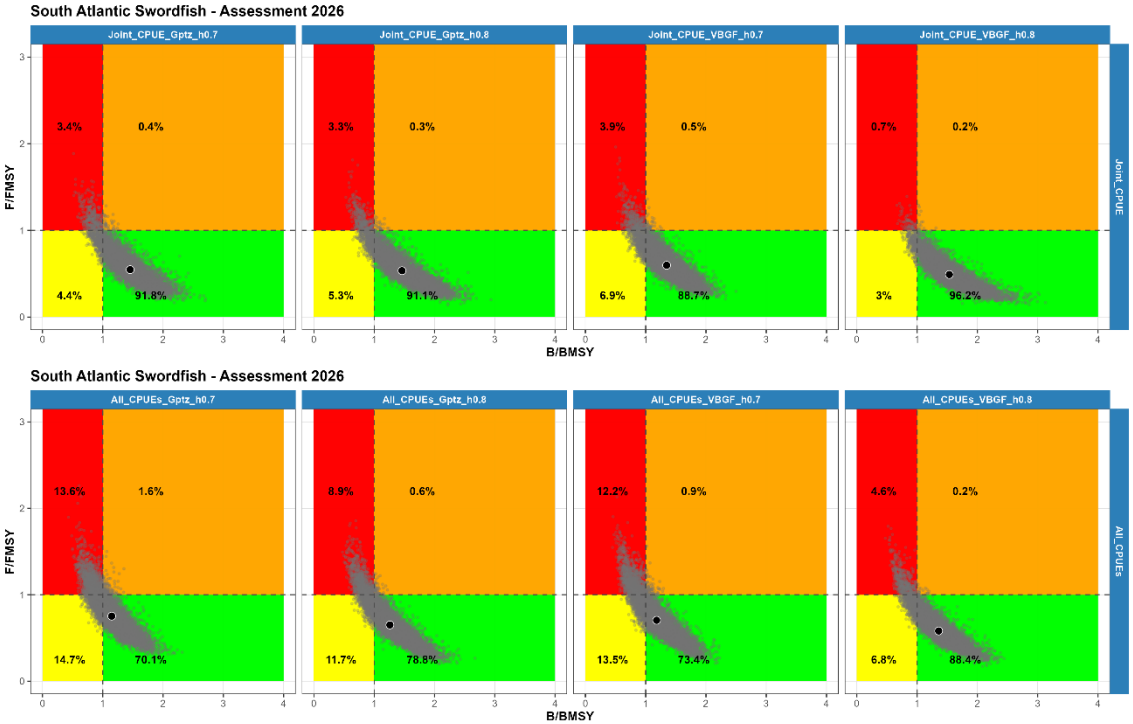


Figure 16. Kobe phase plots showing the joint posterior distributions of stock status ($B/BMSY$ and $F/FMSY$) estimated by JABBA for South Atlantic swordfish under eight alternative model configurations. The upper row corresponds to the Joint_CPUE scenarios, whereas the lower row shows the All_CPUEs scenarios. Columns represent the four biological prior combinations derived from the Age Structured Estimation Model (ASEM): Gompertz ($h = 0.7$), Gompertz ($h = 0.8$), VBGF ($h = 0.7$), and VBGF ($h = 0.8$). Grey points represent posterior MCMC samples, the white point indicates the posterior median estimate, and percentages indicate the posterior probability associated with each Kobe quadrant.

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